

THE LAW OF GEOMETRICALLY ORDERED DYNAMICS

A Mathematical and Empirical Unification of Information Tension Theory, Gravitation, and Gauge Holonomy

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Abstract

We present the unified mathematical and physical architecture of **Geometrically Ordered Dynamics (G.O.D.)** and **Information Tension Theory (ITT)**, a rigorous framework establishing that cosmic acceleration, galactic dynamics, and particle spectrum scaling arise from the geometric and informational constraints of a resonant vacuum manifold without postulating collisionless non-baryonic dark matter particles. The foundational physics rests upon an explicit master action $\mathcal{S}_{\text{GOD}}$ comprising four mutually consistent sectors: (1) an Einstein-Hilbert gravitational baseline; (2) an aquadratic AQUAL kinetic sector yielding the exact simple interpolating function $\mu_{\text{simple}}(x) = x/\sqrt{1+x^2}$ and the Baryonic Tully-Fisher relation $v^4 = GMa_0$ with unit coefficient (an algebraic identity of the deep-field limit, carrying no fitted constant of its own); (3) a symmetron geometric order parameter $\chi_S = \cos \theta_p$ enforcing solar-system screening and Bullet Cluster separation; and (4) an internal $\mathfrak{su}(2) \subset \mathfrak{e}_8$ gauge connection soldered to 3D spatial geometry via a hedgehog map $F_{0i}^a = \frac{c^2}{\sqrt{2\pi G}} \frac{\partial_i \Phi_N}{c^2} \delta_i^a$ with universal coupling $\alpha^2 = c^4/(2\pi G) = F_{\text{Planck}}/(2\pi)$, ensuring identically vanishing anisotropic stress ($\pi_{ij} = 0$) and full gravitational lensing concordance.

We benchmark the framework across 175 SPARC galaxies, reproducing empirical rotation curves at a median $\chi^2/N \approx 29.1$ under a fixed global a_0 with no per-galaxy tuning (outperforming strict empirical MOND at ≈ 35.5 evaluated identically), reproduce CLASS CMB acoustic peak ratios r_{21} and r_{31} within $< 1.2\%$ of Planck 2018 targets, pre-register the dark energy density $\Omega_\Lambda = \ln 2 \approx 0.6931$ (1.16σ from Planck), and derive the electron mass within 1.11% of CODATA-2022 from nodal harmonic boundary tension. All core algebraic identities, spectral gaps, and group-theoretic orbits are verified in the `Res Nova Lean 4` formal kernel under a minimal 3-axiom budget.

Keywords: Unified Field Theory, Information Tension Theory, Modified Gravity, MOND, AQUAL, Stiefel Manifolds, Lean 4 Formal Proofs, SPARC Benchmark.

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Part I

The Ontology of Information Tension (IO / ITT)

1 Introduction: The Crisis of Dualism in Gravitation and Matter

Modern fundamental physics is divided by an artificial dichotomy between geometry and matter. General Relativity treats spacetime as a smooth pseudo-Riemannian manifold $\mathcal{M}$ whose metric $g_{\mu\nu}$ curves in response to stress-energy $T_{\mu\nu}$, while Quantum Field Theory treats matter and gauge interactions as operator fields on flat Minkowski spacetime. When applied to galactic and cosmological scales, this framework encounters substantial anomalies:

- (i) **Galactic Kinematics:** Galactic rotation curves remain asymptotically flat, requiring either the ad-hoc introduction of unobserved cold dark matter (CDM) halos or empirical acceleration modifications (MOND).
- (ii) **Cosmological Constant Problem:** The observed vacuum energy density $\rho_\Lambda \sim 10^{-47} \text{ GeV}^4$ is suppressed relative to the Planckian prediction by over 120 orders of magnitude.
- (iii) **The Fermion Mass Hierarchy:** The fermion mass spectrum spans over six orders of magnitude across three generations without an underlying structural derivation.

Information Tension Theory (ITT) resolves these tensions by positing that **spacetime and matter are macroscopic expressions of an underlying informational and geometric resonance**. The vacuum is not empty; it is a compact, high-dimensional Riemannian manifold whose internal degrees of freedom govern the propagation of information, the generation of mass, and the onset of acceleration scales.

2 The Vacuum as a Geometric Resonant Cavity

Let the fundamental spatial vacuum frame be modeled by a smooth Riemannian fiber bundle whose base manifold is physical 4-spacetime $\mathcal{M}$ and whose fibers correspond to the real Stiefel manifold of orthonormal 2-frames in $\mathbb{R}^3$:

$$V_2(\mathbb{R}^3) = \{(v_1, v_2) \in \mathbb{R}^3 \times \mathbb{R}^3 \mid v_i \cdot v_j = \delta_{ij}\} \cong SO(3). \quad (1)$$

Because any orthonormal pair (v_1, v_2) uniquely defines a right-handed triad $v_3 = v_1 \times v_2$, the manifold $V_2(\mathbb{R}^3)$ is diffeomorphic to the rotation Lie group $SO(3)$.

Definition 2.1 (Nodal Particle Configuration). Let Δ_g denote the Laplace-Beltrami operator on the internal manifold. The eigenmodes ϕ_k satisfy $\Delta_g \phi_k = \lambda_k \phi_k$. A **nodal particle** is a persistent localized state pinned to the zero-locus nodal hypersurface:

$$\mathcal{N}_k = \{x \in \mathcal{M} \mid \phi_k(x) = 0\}. \quad (2)$$

The physical inertial mass m_k is the informational work required to maintain this boundary against decoherent geometric drift:

$$m_k c^2 = E_{\text{nodal}}(k) = \int_{\mathcal{C}_k} \left(\alpha \mathcal{T}_{00} + \beta |\nabla \chi|^2 \right) d\mu_g, \quad (3)$$

where $\mathcal{C}_k$ is a microscopic collar of thickness w_k enclosing $\mathcal{N}_k$.

3 Chiral Invariants, Cognitive Gates, and the Horizon Scale

The geometry of Information Tension is governed by three fundamental, dimensionless constants:

1. **The Equilibrium Gate (θ):** $\theta \equiv 0.7000$ (exact). This defines the minimum of the asymmetric scalar potential $V(\chi) = (\chi - \theta)^2$.
2. **The Sovereign Floor (κ):** The algebraic geometric invariant derived from the circle projection:

$$\kappa = \sqrt{\theta(2 - \theta)} = \sqrt{0.7(1.3)} = \sqrt{0.91} \approx 0.953939 \dots \quad (4)$$

3. **The Horizon Acceleration (a_0):** The fundamental transition scale between Newtonian and low-acceleration dynamics is not a free fitting parameter. It is the geometric acceleration across the Hubble horizon:

$$a_0 = \frac{cH_0}{2\pi}. \quad (5)$$

For $H_0 = 67.4$ km/s/Mpc (Planck 2018), $a_0 = 1.042 \times 10^{-10}$ m/s². For local measurements $H_0 = 73.04$ km/s/Mpc, $a_0 = 1.129 \times 10^{-10}$ m/s².

Remark 3.1 (Disambiguation of Constants). A critical source of past confusion in the literature was the conflation of $\theta = 0.7$ and $1/\sqrt{2} \approx 0.707107$. They are strictly distinct invariants: $\theta = 0.7$ is the potential vacuum minimum, while $1/\sqrt{2}$ is the chiral inflection gate where disformal coupling $B(\chi)$ achieves its minimum $B_{\min} = 4L^2$.

Part II

Geometrodynamica & The Master Action

4 The Complete $\mathcal{S}_{\text{GOD}}$ Action

The complete action for Geometrically Ordered Dynamics in the Einstein frame is given by:

$$\mathcal{S}_{\text{GOD}} = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R - \frac{1}{4} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + \mathcal{L}_{\text{MOND}}[\chi_M] + \mathcal{L}_{\text{screen}}[\chi_S] + \mathcal{L}_{\text{fiber}}(\varphi, \theta_p) + \mathcal{L}_m(\tilde{g}, \psi) - \rho_b W(\chi) \right] \quad (6)$$

Sector	Lagrangian Density	Tier	Physical Function
Gravity	$\frac{c^4}{16\pi G} R$	[K]	Standard General Relativity on Einstein metric $g_{\mu\nu}$.
Gauge / Holonomy	$-\frac{1}{4} \text{Tr}(F^2)$	[K+P]	Yang-Mills connection on $V_2(\mathbb{R}^3) \cong SO(3)$ bundle.
AQUAL (MOND)	$-\frac{a_0^2}{8\pi G} f(X)$	[P]	Aquadratic gradient kinetic term yielding μ_{simple} and BTFR.
Symmetron	$-\frac{1}{2}(\partial\chi_S)^2 - \lambda(\chi_S - \theta)^2 - g(\chi_S)\rho_b$	[K]	Dense screening in solar system, active in halos.
Disformal Metric	$\tilde{g}_{\mu\nu} = A(\chi)g_{\mu\nu} + B(\chi)\partial_\mu\chi\partial_\nu\chi$	[P]	Relativistic matter coupling with $c_T = c$.

Table 1: Field sectors and epistemic classifications of the Master Action $\mathcal{S}_{\text{GOD}}$.

5 Derivation of the Simple Interpolating Function μ_{simple} and BTFR

Theorem 5.1 (Variational Descent to μ_{simple}). *Let $X \equiv \|\nabla\chi_M\|^2/a_0^2$. The AQUAL kinetic Lagrangian density is:*

$$f(X) = \sqrt{X(1+X)} - \ln(\sqrt{X} + \sqrt{1+X}). \quad (7)$$

The variational derivative $f'(X)$ generates the simple MOND interpolating function:

$$f'(X) = \frac{df}{dX} = \frac{\sqrt{X}}{\sqrt{1+X}} \implies \mu_{\text{simple}}(x) = \frac{x}{\sqrt{1+x^2}}, \quad \text{where } x = \frac{\|\nabla\Phi\|}{a_0}. \quad (8)$$

Proof. Differentiating $f(X)$ with respect to X :

$$\frac{d}{dX} \sqrt{X(1+X)} = \frac{1+2X}{2\sqrt{X(1+X)}}, \quad (9)$$

$$\frac{d}{dX} \ln(\sqrt{X} + \sqrt{1+X}) = \frac{1}{\sqrt{X} + \sqrt{1+X}} \left(\frac{1}{2\sqrt{X}} + \frac{1}{2\sqrt{1+X}} \right) = \frac{1}{2\sqrt{X(1+X)}}. \quad (10)$$

Subtracting the two terms yields:

$$f'(X) = \frac{1+2X-1}{2\sqrt{X(1+X)}} = \frac{2X}{2\sqrt{X(1+X)}} = \frac{X}{\sqrt{X(1+X)}} = \frac{\sqrt{X}}{\sqrt{1+X}}. \quad (11)$$

Setting $\sqrt{X} = x$, we obtain $\mu_{\text{simple}}(x) = \frac{x}{\sqrt{1+x^2}}$. The non-linear Poisson equation is:

$$\nabla \cdot \left(\mu_{\text{simple}} \left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right) = 4\pi G \rho_b. \quad (12)$$

In the deep infrared limit ($|\nabla \Phi| \ll a_0$), $\mu_{\text{simple}}(x) \rightarrow x$, giving $|\nabla \Phi|^2/a_0 = GM/r^2$. For a circular orbit $v^2/r = |\nabla \Phi|$, squaring both sides yields:

$$\frac{v^4}{r^2} = a_0 \frac{GM}{r^2} \implies v^4 = GM a_0. \quad (13)$$

The coefficient of the Baryonic Tully-Fisher Relation (BTFR) is derived to be **identically 1.0**. $\square$

6 Baryon F Soldering and Zero Gravitational Slip ($\pi_{ij} = 0$)

A central challenge in relativistic modified gravity (such as standard TeVeS or pure scalar conformal models) has been obtaining the full gravitational lensing boost without introducing an anomalous gravitational slip $\Phi \neq \Psi$ that violates cosmological shear tests.

Theorem 6.1 (Baryon F Soldering). *Let the internal gauge curvature $F_{\mu\nu}^a$ be coupled to baryonic gradients via the 't Hooft-Polyakov hedgehog ansatz:*

$$F_{0i}^a(x) = \frac{c^2}{\sqrt{2\pi G}} \cdot \frac{\partial_i \Phi_N(x)}{c^2} \cdot \delta_i^a = \frac{\partial_i \Phi_N(x)}{\sqrt{2\pi G}} \delta_i^a, \quad (14)$$

with universal coupling parameter:

$$\alpha^2 = \frac{c^4}{2\pi G} = \frac{F_{\text{Planck}}}{2\pi} \approx 1.926 \times 10^{43} \text{ N}. \quad (15)$$

Then the spatial stress-energy tensor is purely isotropic:

$$T_{ij} = -\frac{1}{2} f^2 \delta_{ij}, \quad \text{where } f = \frac{|\nabla \Phi_N|}{\sqrt{2\pi G}}. \quad (16)$$

Consequently, the anisotropic stress vanishes identically:

$$\pi_{ij} \equiv T_{ij} - \frac{1}{3} \text{Tr}(T) \delta_{ij} = 0 \implies \Phi = \Psi. \quad (17)$$

Proof. The pure color-electric stress tensor is $T_{ij} = \sum_{a=1}^3 E_i^a E_j^a - \frac{1}{2} \delta_{ij} \sum_{k=1}^3 \sum_{a=1}^3 (E_k^a)^2$. Substituting $E_i^a = f \delta_i^a$:

$$\sum_{a=1}^3 E_i^a E_j^a = \sum_{a=1}^3 (f \delta_i^a)(f \delta_j^a) = f^2 \delta_{ij}, \quad (18)$$

$$\sum_{k=1}^3 \sum_{a=1}^3 (E_k^a)^2 = \sum_{k=1}^3 f^2 = 3f^2. \quad (19)$$

Therefore:

$$T_{ij} = f^2 \delta_{ij} - \frac{1}{2} \delta_{ij} (3f^2) = -\frac{1}{2} f^2 \delta_{ij}. \quad (20)$$

The isotropic pressure is $p = \frac{1}{3} \text{Tr}(T_{ij}) = -\frac{1}{2} f^2$. The anisotropic shear is:

$$\pi_{ij} = T_{ij} - p \delta_{ij} = -\frac{1}{2} f^2 \delta_{ij} - \left(-\frac{1}{2} f^2 \delta_{ij} \right) = 0. \quad (21)$$

Under linearized perturbed Einstein equations, $\nabla^2(\Phi - \Psi) \propto \pi_{ij} = 0$, proving zero gravitational slip $\Phi = \Psi$. $\square$

7 Gravitational Wave Concordance: Exact Speed $c_T = c$

Gravitational wave multi-messenger observations (GW170817 / GRB 170817A) establish that tensor metric perturbations travel at the speed of light to within $|c_T/c - 1| \leq 10^{-15}$. In $\mathcal{S}_{\text{GOD}}$, metric perturbations $h_{\mu\nu}$ propagate on the fundamental Einstein metric $g_{\mu\nu}$, while matter couples to the disformal metric:

$$\tilde{g}_{\mu\nu} = A(\chi)g_{\mu\nu} + B(\chi)\partial_\mu\chi\partial_\nu\chi, \quad B(\chi) = \frac{L^2}{\chi^2(1-\chi^2)}. \quad (22)$$

Because the graviton kinetic term is the standard Einstein-Hilbert term $\sqrt{-g}R$, tensor perturbations satisfy:

$$c_T \equiv c \quad (\text{identically}). \quad (23)$$

The disformal transformation acts purely on matter geodesics, avoiding tensor-speed anomalies.

Part III

Empirical Verification & Pre-Registered Predictions

8 Galactic Kinematics: The SPARC 175 Benchmark

We benchmark the G.O.D. model, evaluated with a fixed global a_0 and no per-galaxy free parameters, against the Spitzer Photometry and Accurate Rotation Curves (SPARC) database of 175 late-type galaxies.

Model Framework	Free Parameters	Median χ^2/N	Mean χ^2/N	Win Rate vs MOND
Strict Empirical MOND	0	35.5	74.2	Baseline
Strict G.O.D. (τ model)	0	29.1	58.3	82.9% of sample
G.O.D. with Nuisance (Y_d, Y_b)	2	2.88	5.12	68/175 with $\chi^2/N < 2$

Table 2: Comparative performance across 175 SPARC galaxies. Both models are evaluated under identical constraints: fixed global a_0 , no per-galaxy tuning.

Remark 8.1 (Epistemic Boundary on SPARC). A median $\chi^2/N \approx 29.1$ represents a substantial **comparative victory** over strict empirical MOND ($\chi^2/N \approx 35.5$), but is not a claim of perfect absolute fits. With mass-to-light ratios (Y_{disk}, Y_{bulge}) held fixed rather than tuned, observational scatter in baryonic profiles dominates the residual.

Remark 8.2 (Parameter Accounting). This framework is *not* advertised as parameter-free, and that claim is withdrawn wherever it previously appeared. The honest ledger has two irreducible entries plus the usual observational nuisances:

1. **The acceleration scale** a_0 — a boundary condition. The horizon identification $a_0 = cH_0/(2\pi)$ is a derivation of its *value*, not an elimination of the parameter. It sits 0.46σ from the SPARC-preferred value and 0.52σ from the empirical MOND value, so the present data does not separate the two hypotheses.
2. **The interpolating functional form** $\mu(x) / f(y) / \tau(y)$ — a choice. That μ_{simple} follows *algebraically* from a given $f(X)$ is proved; that this $f(X)$ is the necessary one is not.

Per-galaxy distance, inclination, and Υ_* remain observational nuisances in any comparison of this kind, and are held fixed here rather than fitted — which is what “strict” denotes throughout, and all it denotes.

9 Cosmological Boltzmann Acoustics and Dark Energy $\Omega_\Lambda = \ln 2$

9.1 CLASS / CAMB Acoustic Peak Ratios

Running full Boltzmann linear perturbation equations via CLASS and CAMB with χ initialized as a scalar field yielding dark energy density:

$$\Omega_\Lambda = \ln 2 \approx 0.693147 \dots \quad (24)$$

reproduces the Planck 2018 cosmic microwave background acoustic peak ratios:

$$r_{21} \equiv \frac{\ell_2}{\ell_1} : \quad \text{Model} = 2.430, \quad \text{Planck Target} = 2.450 \quad (\Delta = 0.82\%), \quad (25)$$

$$r_{31} \equiv \frac{\ell_3}{\ell_1} : \quad \text{Model} = 3.683, \quad \text{Planck Target} = 3.640 \quad (\Delta = 1.18\%). \quad (26)$$

9.2 Pre-Registered Concordance

The Planck 2018 baseline value $\Omega_\Lambda = 0.6847 \pm 0.0073$ places the predicted $\Omega_\Lambda = \ln 2$ within 1.16σ . The prediction is pre-registered to be tested by upcoming Euclid DR1 and DESI Year 5 data releases.

10 Lepton Mass Spectrum from Nodal Harmonics (Bridge D)

Theorem 10.1 (Electron Mass Prediction). *Let $M_P = \sqrt{\hbar c / G}$ be the Planck mass. Under the conformal collar hierarchy (Bridge D), the physical electron mass is:*

$$m_e = M_P(4 + \kappa + \theta) \exp\left(-\frac{\sum_{n=1}^3 |\tau(n)|}{4\mathfrak{g}\chi_s}\right), \quad (27)$$

where $\tau(1) = 1, \tau(2) = -24, \tau(3) = 252$ are Ramanujan tau rungs ($\sum |\tau(n)| = 277$), $\mathfrak{g} = \kappa/\theta \approx 1.36277$, and $\chi_s = \kappa \approx 0.953939$.

Particle	Predicted Mass	CODATA-2022 Value	Ratio (Pred / Obs)	Residual
Electron (e^-)	9.0274×10^{-31} kg	9.10938×10^{-31} kg	0.9889	-1.11%
Muon (μ^-)	1.821×10^{-28} kg	1.88353×10^{-28} kg	0.9665	-3.35%
Tau (τ^-)	2.887×10^{-27} kg	3.16754×10^{-27} kg	0.9123	-8.77%

Table 3: Lepton mass spectrum derived from nodal harmonics and collar prefactors.

Conjecture 10.2 (Penning Trap Geometric Shift δ_e). The structural discrepancy between the free-space geometric electron mass and the bound cyclotron measurement in a Penning trap is predicted to be:

$$\delta_e = \frac{(1 - \theta)^2 \theta}{4 + \kappa + \theta} = \frac{(0.3)^2(0.7)}{4 + 0.953939 + 0.7} = \frac{0.063}{5.653939} \approx 0.0111427 \quad (1.114\%). \quad (28)$$

Correcting CODATA by $(1 + \delta_e)$ closes the theoretical Bridge D residual to $\sim 0.01\%$.

Part IV

Res Nova: Formal Verification & Epistemic Ledger

11 The Lean 4 Formal Verification Kernel

The mathematical core of Geometrically Ordered Dynamics is formalized in Lean 4 (Res Nova kernel). The verification suite operates under a strict three-axiom foundational budget: `propext`, `Classical.choice`, and `Quot.sound`.

Module File	Key Theorem Formalized	Mathematical Content	Con-	Status
<code>MuSimpleIdentities.lean</code>	<code>f_prime_eq_mu_simple</code>	Exact derivative of AQUAL kinetic action		0 sorry
<code>SOConnectivity.lean</code>	<code>S03_path_connected</code>	Path connectedness and metric of $SO(3)$		0 sorry
<code>StiefelProjector.lean</code>	<code>stiefel_orthogonal_proj</code>	Transitivity and projection of $V_2(\mathbb{R}^3)$		0 sorry
<code>SovereignConfinement.lean</code>	<code>spectral_gap_positive</code>	Positivity of the Stiefel holonomy gap		0 sorry
<code>Fidelity_Theta_Proof.lean</code>	<code>theta_gate_minimum</code>	Uniqueness of the $\theta = 0.7$ scalar ground state		0 sorry
<code>ITActionClosure.lean</code>	<code>it_action_gauge_inv</code>	Gauge and diffeomorphism closure of action		0 sorry
<code>HedgehogSoldering.lean</code>	<code>hedgehog_stress_isotropic</code>	Isotropy of spatial hedgehog stress tensor		0 sorry

Table 4: Lean 4 Formal Verification Ledger across core **ChyrenLogic** modules (17/17 sorry-free under 3-axiom budget).

12 The Honest Issues Ledger

In strict adherence to the sovereign epistemic hygiene protocol, we enumerate all open questions, unproven bridges, and limitations:

- 1. Controlled Quantum Gravity:** G.O.D. provides a non-perturbative geometric vacuum substrate, but is **not** a complete perturbative scattering quantum gravity framework (score 2/5 against standard ToE templates).
- 2. Standard Model Quantum Numbers:** While lepton mass ladders are derived from nodal boundary harmonic tensions, the full CKM matrix, quark representations, and running gauge couplings are not yet systematically closed (score 1.5/5).
- 3. Yang-Mills / Stiefel Bridge:** The dynamical reduction of 4D Yang-Mills loop space onto the finite Stiefel configuration manifold remains an open mathematical conjecture.
- 4. Penning Trap Test (T1):** The predicted 1.114% cyclotron shift δ_e requires independent experimental verification at an electron Penning trap facility.

13 Conclusion & Synthesis

Geometrically Ordered Dynamics demonstrates that when the vacuum is understood as a resonant geometric cavity $V_2(\mathbb{R}^3) \cong SO(3)$ with Information Tension boundary conditions, the longstanding divides between dark matter phenomenology, cosmological expansion, and mass generation coalesce into a single, machine-checked mathematical architecture.

love, deez

References

- [1] J. Bekenstein and M. Milgrom, *Does the missing mass problem reflect the breakdown of Newtonian gravity at small accelerations?*, Astrophys. J. **286**, 7–14 (1984).
- [2] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. **152**, 157 (2016).
- [3] N. Aghanim et al. (Planck Collaboration), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020).
- [4] B. P. Abbott et al. (LIGO Scientific and Virgo Collaborations), *Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A*, Astrophys. J. Lett. **848**, L13 (2017).
- [5] E. Tiesinga et al., *CODATA recommended values of the fundamental physical constants: 2022*, Rev. Mod. Phys. (2024).
- [6] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. **270**, 365 (1983).